Explained variation and human importance

Response to the reviewer

We thank the reviewer for asking how much variation the human-impact and climate predictors explain, and whether model fit covaries with the estimated importance of human impact.

We added a direct commonality decomposition for every available H1 model. Unique human, unique climate, shared, and total adjusted R^2 fractions come from `Var.part`. Hierarchical individual importance remains a separate quantity from `Hier.part`. This follows the rdacca.hp contract and Lai et al. (2022).

For each model, total adjusted R^2 equals unique human plus unique climate plus shared explained variation. Adding unexplained variation reconstructs one. Signed adjusted fractions are authoritative for decomposition accounting and for the correlation diagnostic below. The main ecological figures instead use zero-truncated and renormalised compositions of positive hierarchical contributions; these two presentations are not interchangeable. These are joint multivariate models of ten pollen-derived properties, not ten independent fits.

Across the four H1 analyses, 1565 of 2679 model records had complete decompositions. No available model exceeded the 0.001 accounting tolerance. For spatial SPD, 1156 decompositions had a median adjusted R^2 of 0.355 (interquartile range 0.231 to 0.482). We show record-level distributions rather than sums because sums scale directly with the number of models in each group.

Table 1: Availability, accounting, and signed-component audit for H1.

Analysis	Models	Available	Invalid	Neg. human	Neg. climate	Neg. shared
spatial_events	1262	263	0	56	31	74
spatial_spd	1262	1156	0	428	108	347
temporal_events	85	76	0	13	0	14
temporal_spd	70	70	0	13	1	22

Three independent figure products

The first figure shows the distribution of signed adjusted R^2 values within each continent and climate-zone combination. Points are individual spatial SPD models, boxes show medians and interquartile ranges, and violins show distribution shape when at least three varying values are available. Finite negative adjusted R^2 values are retained.

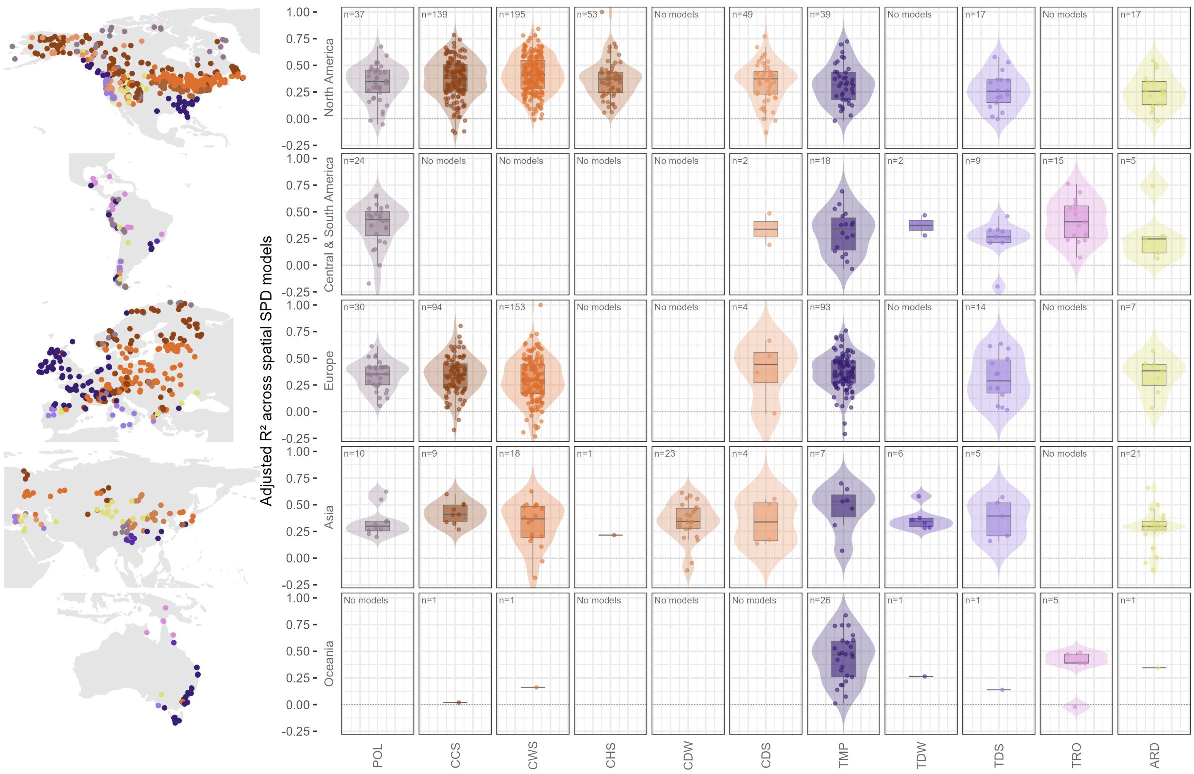

Figure 1: Adjusted R-squared distributions by continent and climate zone. Points are individual models; boxes show medians and interquartile ranges.

The correlation figures are supplementary diagnostics and use the untruncated signed human hierarchical contribution only; no zero-truncated composition is shown. Across 1118 eligible spatial SPD models, Spearman = 0.022 and Pearson r = -0.021.

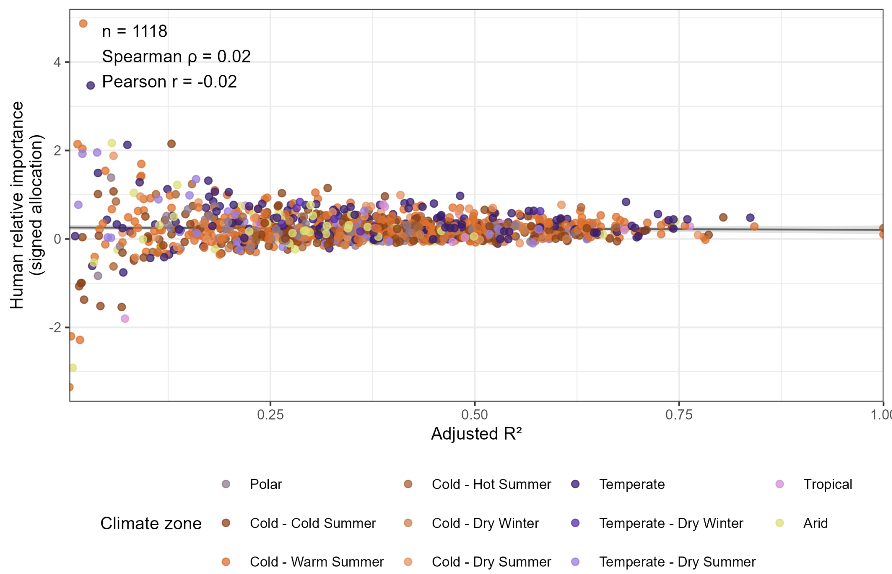

Figure 2: Adjusted R-squared and the untruncated signed human hierarchical contribution across eligible spatial SPD models. This is a supplementary decomposition diagnostic, not the zero-truncated composition used in the main ecological figures.

The grid uses five continent rows and eleven climate-zone columns. Populated panels report n and Spearman . Statistics and trend lines require at least three observations and non-zero variation. Absent combinations remain explicit empty cells.

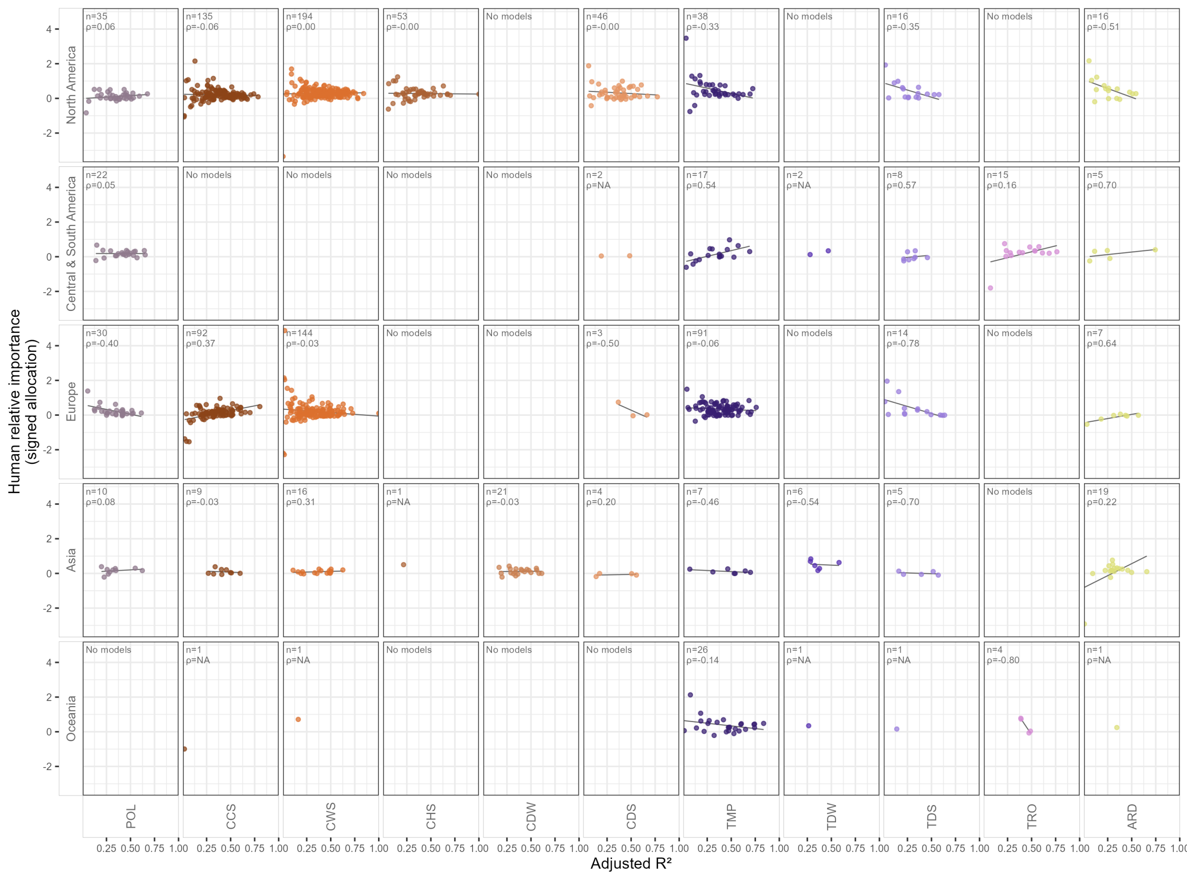

Figure 3: Adjusted R-squared and the untruncated signed human hierarchical contribution by continent and climate zone. This is a supplementary decomposition diagnostic.

All model values, distribution summaries, audits, and signed correlation statistics are exported under `Outputs/Tables/Diagnostics/HVarPart`. The validation script independently reconstructs every distribution summary, model count, Spearman coefficient, and Pearson coefficient.